Class Meeting Handout #18

Barry Linkletter

This document was produced using the LaTeX typesetting language with the Tufte-handout document class. Chemical diagrams were created in *ChemDoodle* Plots were typeset using *MatPlotLib* and *Python* tools. Diagrams were created and edited using *Affinity Designer*.

We can investigate the changes that occur in a bond during a reaction using reaction kinetics and isotopic substitution. Today we will explore the theory of these kinetic isotope effects.[1]

[1] Read chapter 8.1.

Kinetic Isotope Effects

The chemistry of the elements is defined by how many protons are in a nucleus and the quantum world that controls the electrons around it. Neutrons are required to have a stable nucleus for all elements heavier than hydrogen. How many neutrons? It turns out that there are options. Six protons and six neutrons is the stable element carbon-12. This element has been assigned the mass of exactly twelve grams per mole.[2] Six protons and seven neutrons is the stable element carbon-13. It is 1 % of carbon on our planet. It has the same number of protons and electrons as carbon-12, and will have identical chemistry. But it is heavier, and that has consequences.

[2] It's the standard.

Exercise: Primary Kinetic Isotope Effect – 50 minutes

In your required reading, you will have considered these ideas already. Let us now discuss them as a class. Much of this will require us to recall knowledge from spectroscopy. Will the review never end in this course?

Useful formulas

$$E_V = (V + \tfrac{1}{2})h\nu$$
$$E = h\nu$$
$$\nu = \bar{\nu}c$$
$$\nu = \frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$$
$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Useful constants

$$h = 6.626 \times 10^{-34}\ \mathrm{J\,s}$$
$$c = 2.998 \times 10^{10}\ \mathrm{cm\,s^{-1}}$$
$$N = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$$

1. Draw a C – H bond. Draw a diagram of a harmonic oscillator for that bond and indicate several energy levels. Where is the lowest energy level?

2. Using the IR spectrum provided in Figure 1 on the next page, estimate the stretching frequency of the C – H bond and use Hooke's law to calculate the force constant, k. Use dynes as the unit of force and cm for distance.[3]

3. The force constant is due to the electronic nature of the bond. It will not change with isotopic substitution. Now, calculate the stretching frequency of the corresponding C – D bond. Compare your result with the experimental evidence presented in Figure 2 on the next page.

4. Calculate the energy of the $v = 0$ vibrational energy level in both the cases: a C – H and a C – D bond. Draw these energy levels on the harmonic oscillator diagram for comparison.

5. Draw a reaction coordinate diagram the expresses the cleavage of the C – H bond. Add the cleavage of a C – D bond to the diagram.[4] With this diagram in mind, calculate the relative rate for C – H cleavage compared to C – D cleavage, k_H/k_D. This value will be the maximum possible value for the primary kinetic isotope effect for reactions involving the C – H bond.

[3] A dyne is defined as $\mathrm{g\,cm\,s^{-2}}$

[4] The separated atoms will have no vibrational energy levels (the force constant is zero).

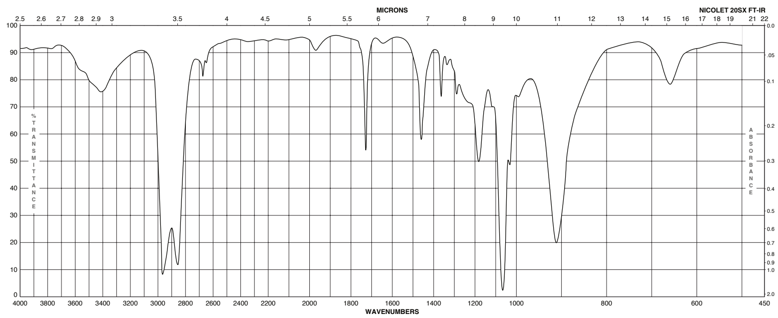

Figure 1: FTIR spectrum of tetrahydrofuran. From the Spectral Database for Organic Compounds at https://sdbs.db.aist.go.jp/sdbs/cgi-bin/landingpage?sdbsno=497 ⟵

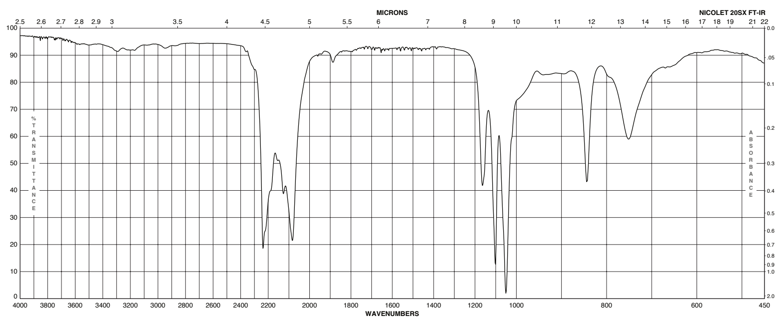

Figure 2: FTIR spectrum of deuterated tetrahydrofuran-d_8. From the Sigma-Aldrich catalogue at https://www.sigmaaldrich.com/CA/en/product/aldrich/184314 ⟵

6. The above steps assumed that the force constant changed from the calculated value in the starting materials to a value of zero in the products where the bond is broken. Imagine a transition state where the force constant is 50 % compared to the ground state. What is the value of the PKIE?

7. In the secondary kinetic isotope effect, the energy difference results from changes in the force constants of bond bending vibrations and is not due to the bonds breaking. One example is the change in hybridization as a carbon is converted from sp^3 TO sp^2 in a reaction. Use the data in Figure 8.6 of the textbook[5] and calculate the value of k_H/k_D.[6]

8. Do you believe that the TS of a reaction will have this hybridization change fully completed? Is the value you just calculated best describes as an an estimate of the maximum value?

9. Consider E1 and E2 elimination mechanisms. Predict the magnitude of the kinetic isotope effect with various substitutions in the E1 and the E2 eliminations for t-butyl bromide. Give correct reaction conditions that favour each mechanism.[7]

[5] See page 429 of the textbook. You could also consider the data presented Table 4 (page 28) of the kinetic isotope tutorial document available on the Moodle site.

[6] If you were performing your calculations using a *Python* notebook, then you might already have some lines of code that will answer this question with little or no modification. I will demonstrate all the calculations in this exercise using *Python* as my calculator. Observe how it results in clear documentation of my mathematical approach and also a tool that I can use again and again.

[7] The review never ends. Recall this example from your sophomore organic chemistry course. It's a classic.

Learning Goals

Through the suggested textbook reading, problems and participating in this class discussion, students will...

- have reviewed vibrational spectroscopy (a bit)
- now understand that electronic energy is not the only energy that affects reaction rates and equilibrium.
- be able to express the definitions involved in isotope effects
- understand the theory of isotope effects and be able to apply it in calculations.
- be able to calculate the maximum theoretical isotope effect using bond vibration frequencies.
- be able to use the magnitude of primary and secondary isotope effects to interpret how a bond changes during a reaction.
- be able to predict the magnitude of an isotope effect in an experimental design and interpret experimental results.
- **Computer Skill**: You will have again observed how much better a *Python* notebook serves you in scientific calculations compared to a calculator.

A Look Ahead

We have been introduced to the theory of kinetic isotope effects. In the next class meeting we will examine some examples and practice using isotope effects to test hypotheses for mechanisms.

Also, I think there's a test coming up. You should check the syllabus for the date.